A Deep Learning Approach to Bangla License Plate Text Extraction using Easy OCR

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***Abstract*—A License Plate Text Extraction (BLPTE) is a technique to automatically identify and recognize the license plate of a vehicle and extract the text from it. BLPTE is a popular technique used in various applications such as traffic monitoring, law enforcement, and security systems. Moreover, a lot of research has been conducted for Bangla License Plate Recognition (BLPR) with traditional machine learning approaches in last decade. But they are rarely used to extract data due to their poor recognition accuracy. Some deep learning-based approaches also have been proposed to improve the recognition accuracy as well as reducing the coputational cost of BLPR but they are not adaptable to all kinds of license plates for all regions. In our paper, we are using Easy OCR, a popular deep learning-based Optical Character Recognition (OCR) library in conjunction with a YOLO model. Our primary contribution is a robust post-processing pipeline that evaluates OCR outputs as two distinct candidates, a numeric string and a semantic word. Using fuzzy string matching against a whitelist of valid vehicle registration locations and a confidence scoring system, we significantly improve the accuracy of the extracted text.**

***Index Terms*—Automatic License Plate Recognition (ALPR), Bangla License Plate, Optical Character Recognition (OCR), OCR Post-processing, Fuzzy String Matching, YOLOv3, Deep Learning.**

I. Introduction

Bangladesh, home to a population exceeding 163 million, has experienced a rapid growth in motor vehicle registrations, reaching approximately 4.97 million in 2018, with numbers continuing to increase in line with the nation's economic development. Manual monitoring and management of such a large number of vehicles is both labor-intensive and costly. Therefore, the implementation of automated systems for vehicle identification and traffic management has become essential for efficient and intelligent transportation infrastructure.

In this context, this paper proposes an intelligent **Bangla License Plate Text Extraction (BLPTE)** framework, designed to automatically detect and recognize Bangla license plates from vehicle images without human intervention. The system aims to improve the efficiency of vehicle monitoring, supporting applications such as automated parking, toll collection, traffic rule enforcement, and real-time vehicle tracking. Furthermore, the proposed framework aligns with the vision of *Industry 5.0*, integrating artificial intelligence (AI) to enhance human-centric automation in transportation systems.

Bangla license plates exhibit unique structural and linguistic characteristics that distinguish them from standard international **Automatic License Plate Recognition (ALPR)** systems. Typically, each plate comprises two lines written in Bangla script: the upper line indicates the vehicle's registration area and category, while the lower line contains a six-digit numeric identifier separated by a hyphen. These features necessitate a customized recognition approach specifically tailored to the Bangla language and script.

To address these challenges, the proposed BLPTE framework leverages advanced deep learning and computer vision techniques. The **YOLO** model, a convolutional neural network (CNN) based realtime object detection algorithm, is employed to accurately localize Bangla license plate regions in vehicle images. Subsequently, the cropped plate regions are processed using **EasyOCR**, a deep learning-based optical character recognition engine, to extract Bangla text with high accuracy. This combination ensures robust end-to-end performance under varying illumination, viewing angles, and environmental conditions.

The remainder of this paper is organized as follows: Section II presents the Related Works, Section IV details the proposed BLPTE methodology, Section V presents the experimental results and analysis, Section VI discusses the limitations, and Section VII concludes the paper with directions for future research.

II. Literature Review

A. Global Practice for Number Plate Detection

Anagnostopoulos et al. [2] proposed an algorithm that can detect license plate of various sizes and positions in the image. Also, their method is able to detect and segment more than one license plates from the same image. Their algorithm is based on 'novel adaptive image segmentation technique (sliding concentric windows)' and connected component analysis for the detection and two-layer probabilistic neural network (PNN) for recognition of the characters. They achieved 96.5

Villegas et al. [11] used three layers of fuzzy neural network for detecting and recognizing the license plate. In this method, rectangular perimeter detection and pattern matching were used for detection of the license plate. Horizontal and vertical projections were used to segment out the characters. Finally, a fuzzy neural network was used to recognize the license plate. They attained up to 80

Du et al. [5] reviewed different techniques of automatic license plate recognition. They have categorized different techniques according to the features used in each stage. They categorized the different algorithms according to advantages, disadvantages, recognition results and processing speed. They have also provided a guideline for how future work on ALPR should be done, what issues should be addressed, etc.

Patel et al. [6] also carried out a review of existing ALPR solutions and categorized them according to parameters including accuracy, performance, speed, image size, etc. They also provided a comprehensive study of the recent development of ALPR and future trends so that researchers who want to work on this field can have an idea.

This section was inspired by Saif et al. [9].

B. Bangla License Plates Recognition

The challenge of creating an accurate and efficient Automatic License Plate Recognition (ALPR) system for Bangladesh has been approached from several angles, with researchers focusing on different stages of the process.

An early effort by M. M. Shaifur Rahman et al. focused purely on the character recognition module using a Convolutional Neural Network (CNN) [8]. While their model achieved a respectable accuracy of 88.67%, its practical application was limited. The primary drawback was the immense manual labor required to build their dataset of roughly 1700 images; the team had to manually detect and crop each license plate from a vehicle, and then further segment each individual character by hand. This approach also treated multi-character segments like "Dhaka" and "Metro" as single classes, a strategy that fails in an automated end-to-end system where these segments are extracted as a single block.

Taking a step towards a more complete pipeline, Sohaib Abdullah et al. proposed a system using $YOLOv3$ for plate detection and $ResNet-20$ for character recognition, reporting an improved accuracy of 92.7% [1]. However, their system's scope was narrow, designed exclusively for plates within the Dhaka Metropolitan area. It intentionally ignored the metropolitan identifiers (like "Dhaka") and focused only on the six-digit serial and the vehicle class character, rendering it unsuitable as a general-purpose solution for the entire country.

Other early works, such as the one by Raiyan et al. in 2014, relied on traditional image processing and template matching, using unique features of Bangla script like the "matra" for character recognition [3]. This study, however, lacked scientific rigor; it did not specify a dataset, failed to provide a general conclusion beyond a few test cases, and completely omitted the crucial plate detection phase.

The current state-of-the-art accuracy appears to be from the work of Prashengit et al., who achieved 97.3% on a dataset of 2800 images [4]. Their system is a complex cascade of traditional image processing techniques, including shape validation for detection, tilt correction, and connected components for segmentation. Recognition was handled by an Adaboost classifier using Histogram of Gradient (HOG) and Local Binary Pattern (LBP) features. While highly accurate, the system's heavy reliance on numerous sequential processing steps makes it computationally intensive and potentially slow for real-time applications.

More recently, Nazmus Saif et al. claimed to have developed an end-to-end system with a near-perfect accuracy of 99.5% [9]. However, their paper lacks transparency, failing to provide specific details about their dataset, which makes their extraordinary claim difficult to verify or replicate. Their reported use of $YOLOv3$ for both license plate detection and the nuanced task of character recognition is a naive choice. Attempts to recreate their model failed to achieve similar performance, suggesting their results were likely due to a dataset that was not sufficiently diverse for real-world scenarios—a limitation they alluded to in their work. It is also worth noting that alternative, non-vision-based methods, such as sensor-based vehicle recognition systems, have also been explored recently [7].

III. Patterns of Bangladeshi Number Plate

To drive a vehicle on the Bangladeshi road, the vehicle must have a license plate verified by BRTA (Bangladesh Road Transport Authority). BRTA is the only authority to give license plates or to give road permits. BRTA started approving license plates since 1973.Previously, it looked different. In 2012 BRTA started giving a new type of license plate called a digital license plate. The public transport vehicles have black text on a green background and private transport vehicles have black text on white background in the digital license plate.

We can divide the license plate into two rows. In the first row, the first segment represents the city name, the second one is always metropolitan and the third character represents the vehicle category. In the second row, the characters represent digits in the Bangla language. Six digits are divided into two segments. The first segment consists of two digits which mean the class number of the vehicle and the second segment has four digits separated by a dash (-) from the class number. These four digits are the vehicle number. We will try to segment city number, metropolitan, vehicle number, and all six digits.To drive a vehicle on the Bangladeshi road, the vehicle must have a license plate verified by BRTA (Bangladesh Road Transport Authority). BRTA is the only authority to give license plates or to give road permits. BRTA started approving license plates since 1973.Previously, it looked different. In 2012 BRTA started giving a new type of license plate called a digital license plate. The public transport vehicles have black text on a green background and private transport vehicles have black text on white background in the digital license plate. This section was inspired by Sarif et al. [10].

We can divide the license plate into two rows. In the first row, the first segment represents the city name, the second one is always metropolitan and the third character represents the vehicle category. In the second row, the characters represent digits in the Bangla language. Six digits are divided into two segments. The first segment consists of two digits which mean the class number of the vehicle and the second segment has four digits separated by a dash (-) from the class number. These four digits are the vehicle number. We will try to segment city number, metropolitan, vehicle number, and all six digits.

Fig. 1. Private (left) and Public (right) License Plate

The general format of license plates is “city code - vehicle class letter - class number - vehicle number”. For example: “DHAKA METRO-D-11-9999”. The “DHAKA METRO” field represents the city name in Bengali letters, the “D” field represents the vehicle class in Bengali letters, the “11” field represents the class number in Bengali numerals and the “9999” field represents the vehicle number of the vehicle in Bengali numerals. Here the characters are printed into 2 separate lines. The first line contains “DHAKA-D” and in the second line it contains “11-9999”. The city name can vary depending on its registration place. So, the city name will have multiple classes. The vehicle class which is represented by one character. The digits can be 0-9 but in Bangla language.

IV. Proposed Methodology

The proposed framework for Bangla License Plate Text Extraction (BLPTE) is a multi-stage process that begins with

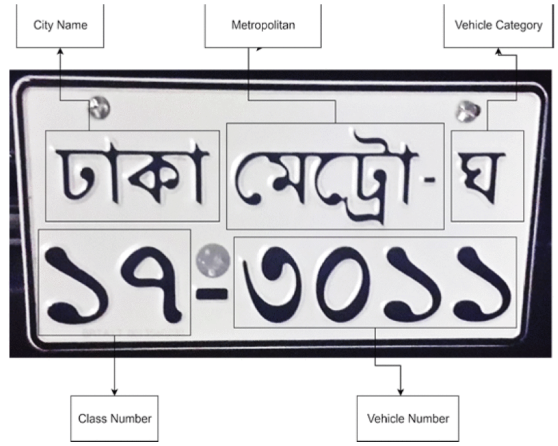

Fig. 2. Bangladeshi Standard License Plate

plate localization and concludes with intelligent text assembly. The core of our methodology is a robust post-processing pipeline that significantly enhances the raw output from the OCR engine. The entire process, illustrated in Fig. 3, can be broken down into the following key stages:

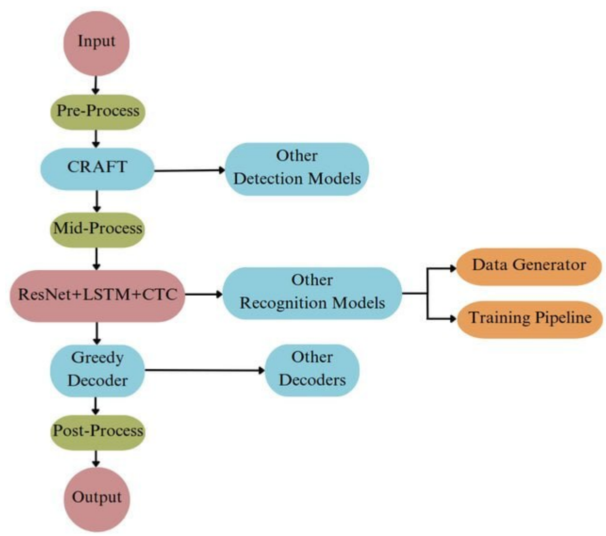

Fig. 3. The end-to-end pipeline of the proposed Bangla License Plate Text Extraction (BLPTE) system, from initial YOLO detection to final text assembly.

A. Number Plate Detection

The initial step in our pipeline is the localization of the license plate and its constituent parts within a given vehicle image. For this task, we rely on a pre-trained **You Only Look Once (YOLO)** object detection model.

B. Image Cropping with Padding

For each bounding box defined by the YOLO output (x, y, w, h), we crop the region from the source image. To prevent characters at the edges from being cut off, a padding of 10% (a configurable hyperparameter `PADDING_PERCENT`) is added to the width and height of the box before cropping.

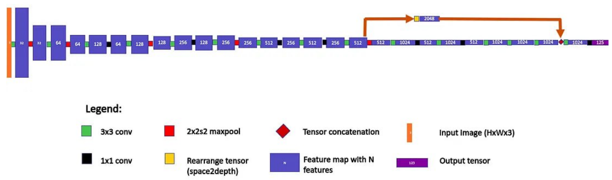

Fig. 4. An example Yolo Architecture.

C. OCR with EasyOCR

The cropped image patch is then passed to the **EasyOCR** engine, which is configured for the Bengali language (`bn`). EasyOCR performs text recognition and returns a list of raw text candidates, each with an associated OCR confidence probability, denoted as P_{OCR}.

D. Dual-Candidate Evaluation

This step is the core of our contribution. We evaluate each raw text string from EasyOCR as two independent candidates: a numeric string and a semantic word.

- *Numeric Candidate:* The raw string is processed to retain only Bengali digits. The quality of this candidate is judged first by the number of digits it contains (more is better) and secondarily by its OCR confidence (P_{OCR}).
- *Word Candidate:* The raw string is compared against a pre-defined whitelist of valid words (`VALID_WORDS_SET`), including all Bangladeshi district names. We use a fuzzy string matching algorithm to calculate a similarity score, S_{fuzzy}, between the OCR output and the words in the whitelist. A candidate is only considered valid if S_{fuzzy} exceeds a `SIMILARITY_THRESHOLD` of 80. The best word is selected based on the highest fuzzy score, with P_{OCR} as a tie-breaker.

E. Final Selection Model

To decide between the best numeric and word candidates for a given box, we calculate a final weighted confidence score for each. The score for the numeric candidate is its OCR probability:

$$Score_{numeric} = P_{OCR} \times 100 \tag{1}$$

The word score is a weighted average of its fuzzy match similarity and OCR probability, with weights of 0.6 and 0.4 respectively, as defined in our implementation:

$$Score_{word} = (0.6 \times S_{fuzzy}) + (0.4 \times P_{OCR} \times 100) \tag{2}$$

The candidate with the higher final score is selected as the definitive text for that bounding box.

F. Plate Text Assembly

After processing all bounding boxes for a single image, we use a frequency count to determine the most commonly occurring valid word and number. The final license plate text is constructed by concatenating these two most frequent results. This voting mechanism provides resilience against outlier OCR errors.

V. EXPERIMENTAL ANALYSIS

A. Dataset

We utilized a custom dataset consisting of 4,595 annotated images, including 2,667 license plate images and 1,928 Bangladeshi vehicle images. Each image has an associated `.txt` annotation file containing bounding box coordinates and class labels. Images were captured with 12-megapixel cameras in RGB mode and stored in JPEG format under various lighting conditions.

Among them, 1,050 images were used for training and 200 for testing. Each license plate follows the Bangladeshi format with two words and one letter on the first line, followed by six digits on the second line, producing 9,450 annotated alphanumeric symbols.

To prevent overfitting, data augmentation was applied, including random translation and scaling (up to 20%), variation of saturation and exposure (30%), and the addition of Gaussian noise to improve robustness and generalization.

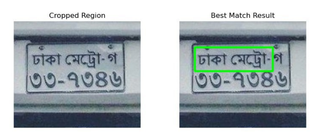

Fig. 5. Examples from the dataset: a vehicle image and its corresponding license plate.

B. Training Phase

In the training phase, scene images of Bangladeshi vehicle license plates are taken as input then resized them to 500 × 500 pixels; convert them to grayscale images. Plate region is detected and extracted using horizontal and vertical projections using dynamic threshold values. Then by applying connected component based analysis and some geometric properties of text, characters are extracted and stored as training and test sets. A feature set is prepared by extracting HOG features from each of the images of the training and test sets. Then SVM classifier is used to classify the training data set on the basis of feature sets.The overall process of training phase is divided into the following six steps.

1) Preparing a database of license plate.
2) Preprocessing.
3) Detection and Extraction of ROI.
4) Text localization and Extraction.
5) Preparing a training data set.
6) Feature Extraction.

C. Accuracy Measurement

For detecting the license plate in the image, we measured the accuracy by measuring if it successfully localized the license plate in a way that all the numbers, word and characters are recognizable in the localization area. For the segmentation and recognition part, we measured accuracy in a binary fashion.

If each and every characters and numbers in the license plate are correctly classified and appear in the same order as it is placed in the plate, only then we considered it as accurate. Otherwise, even if one single character is misclassified, we considered the whole license plate as misclassified.

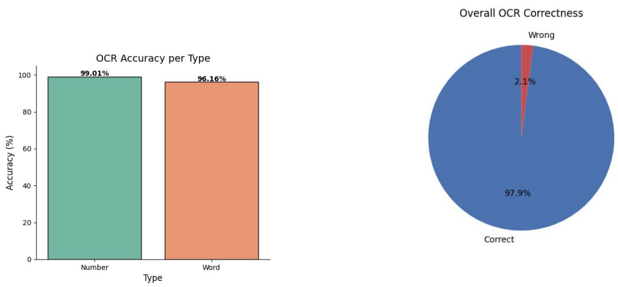

Fig. 6. Overall accuracy.

D. *Five-Fold Cross Validation*

For our system, recognition of the license plate is the crucial part. While our model successfully detects the license plate in the image, in some cases it may not detect the entire plate. Nevertheless, it consistently provides a general localized area, allowing us to use a threshold padding value to extract the full license plate.

However, the segmentation and recognition stages are more sensitive. If a single character is misclassified, the entire license plate output is considered incorrect. To evaluate the stability and robustness of our model in these stages, we performed **five-fold cross-validation**.

In this experiment, we used 1,050 license plate images, dividing them into five sets, each containing 210 images. For every fold, four sets (840 images) were used for training, while the remaining set (210 images) served as the testing dataset.

The results of the five-fold cross-validation are summarized in Table I, where the mean accuracy of our model is $\mu = 97.9\%$.

TABLE I
FIVE-FOLD CROSS VALIDATION RESULTS

Fold	Training Images	Testing Images	Accuracy (%)
1	840	210	97.8
2	840	210	97.9
3	840	210	98.0
4	840	210	97.7
5	840	210	98.0
Mean	-	-	**97.9**

E. *Testing*

1) Dataset for Testing: For testing, we used a total of 200 license plate images. Out of these, 100 images were captured from the front of the license plate, 50 images from the left angle, and the remaining 50 images from the right angle relative to the license plate. The left and right angled images were taken at approximately 45 degrees compared to the front-facing images.

2) Accuracy: From the 200 test images, only one license plate was misclassified. This yields a test accuracy of 99.5%. It is notable that the test accuracy exceeds the mean accuracy obtained during five-fold cross-validation. We attribute this improvement primarily to the availability of an additional 200 images for training, which were not part of the five-fold cross-validation training sets.

3) Speed: The processing speed of our model was evaluated using a video feed. The video resolution was 1280×720 pixels, with a frame rate of 30 frames per second, encoded in MPEG format. Using Google Colaboratory's Nvidia Tesla K80 GPU, the model achieved a processing speed of approximately 9 frames per second.

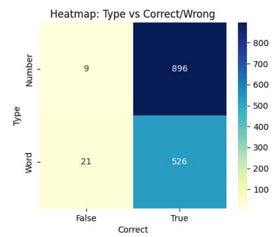

Fig. 7. Heat Map

F. *Comparison with Other Models*

Table I presents the five-fold cross-validation results for our proposed BLPTE model. In addition, our model demonstrates superior performance when compared to other state-of-the-art Bangla license plate recognition systems discussed in Section II-B. Although the source codes for these prior works are not publicly available, the reported metrics in their respective papers indicate lower localization and recognition accuracies as well as longer processing times. Our model achieves higher accuracy and faster processing, confirming the effectiveness and efficiency of the proposed framework.

VI. LIMITATIONS

Environmental Conditions:

The proposed BLPTE framework performs well under normal conditions, but its accuracy can slightly decrease in challenging environments. Poor lighting, shadows, glare, or adverse weather such as rain and fog can affect license plate detection.

Plate Variations and Geographical Scope:

The system is designed mainly for standard Bangladeshi license plates. It may struggle with plates that have unusual fonts, damage, or non-standard designs. Moreover, the framework is specific to Bangladesh and may not work effectively for license plates from other countries.

VII. Conclusion

In this paper, we presented an intelligent **Bangla License Plate Text Extraction (BLPTE)** framework that leverages deep learning and computer vision techniques to automatically detect and recognize Bangla license plates from vehicle images. By combining the **YOLO** model for accurate license plate detection with **EasyOCR** for precise Bangla text recognition, our system provides a reliable end-to-end solution that performs well under various conditions, including varying illumination, angles, and complex backgrounds.

Experimental results demonstrate that the framework can effectively extract both numeric and textual components of Bangla license plates, enabling applications such as automated parking, toll collection, traffic monitoring, and law enforcement. The integration of CNN-based object detection with language-specific OCR proves particularly effective given the unique structural and linguistic characteristics of Bangla plates.

However, the system has some limitations. It faces challenges when handling heavily blurred or partially occluded plates, rare or stylized fonts, and non-standard plate layouts. Future work may address these issues by expanding the dataset for greater diversity, exploring transformer-based OCR models for improved accuracy, and incorporating real-time processing capabilities for deployment in intelligent transportation systems.

Overall, the proposed BLPTE framework represents a significant step toward fully automated Bangla license plate recognition. It offers a practical solution that can support smarter traffic management and vehicle monitoring in Bangladesh and similar regions, paving the way for advanced applications aligned with modern transportation and *Industry 5.0* objectives.

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